

AFL Data Dictionary & Prediction Target Definitions

1 · Analytical tables — grain, keys and joins

Table	One row represents	Primary key	Rows	Joins to
matches	one MATCH	match_id	7,904	—
player_games	one PLAYER-MATCH	(match_id, player_id)	274,079	matches · players · teams
player_seasons	one PLAYER-SEASON-PHASE	(player_id, year, team, is_finals)	25,481	players · teams
players	one PLAYER	player_id	2,843	—
teams	one FRANCHISE	team_id	19	referenced by matches & player_games
match_features_v2.0.0	one MATCH (model-ready)	match_id	5,666	matches

Derived keys. Neither **match_id** nor **team_id** exists in the raw files. **match_id** is built from (match_date, team-pair); **team_id** is minted so tables join on a stable integer rather than a team name that appears in 40 spellings. Join integrity: **0.000%** of player-game rows fail to link — but only after the naming repair in §5. Duplicate player names exist: never join players on name.

2 · Match-level targets — grain: one row per match_id

Target	Type	Formula	Base rate (1997+)	Status
home_win	binary classification	1[home_score > away_score] (draws → 0)	0.5840	PRIMARY
margin	regression	home_score − away_score (+ = home win)	mean +8.09, sd 42.4	secondary
result	3-class	HOME_WIN / AWAY_WIN / DRAW	draw = 0.83%	reference
home_win_or_draw	binary	1[home_score ≥ away_score]	0.5923	draw audit

Framing decision — classification, not regression. The stakeholder asked "who wins", which classification answers directly and with a calibrated probability the chat agent can quote ("Geelong ≈ 68%"); a predicted margin of +12 carries no stated uncertainty. Draws are only 0.83% of matches (~1 in 120) — too rare for a third class, and AFL tipping convention scores a draw as correct for either side, so draws are folded into home_win = 0 (an explicit, documented compromise). Classification is also immune to blowout distortion: a 100-point thrashing and a 1-point win are both simply "a win". Margin is retained as a secondary target because it is information-rich and feeds Elo and ladder percentage; it converts to P(home win) via $\Phi(\hat{\mu}/\sigma)$, $\sigma \approx 42.4$. **margin is recomputed from the scores in the ETL — the raw column is wrong-signed on some rows, and a wrong-signed target silently inverts the label.**

3 · Player-level targets — grain: one row per (match_id, player_id)

Target	Type	Formula	Available	Status
match_impact_score (MIS)	continuous	weighted 13-statistic composite (below)	1999+	PRIMARY
is_top_mis	binary	1[MIS = max over match]	1999+	primary (binary)
is_top_disposal_getter	binary	1[disposals = max over match]; disposals = kicks + handballs	1983+	top-player v1
is_top_goal_kicker	binary	1[goals = max over match AND goals > 0]	1983+	top-player v2
season_goals	count / regression	Σ goals over season (Coleman Medal target)	1983+	top-player v2b
fantasy_points	continuous	3k + 2h + 3m + 4t + 6g + 1b + 1ho + 1ff − 3fa	1987+	external benchmark

Match Impact Score — exact formula

MIS = 4.0·goals + 3.0·goal_assist + 2.5·clearances + 2.0·tackles + 1.5·marks + 1.5·inside_50s + 1.0·disposals + 1.0·contested_possessions + 1.0·rebound_50s + 0.5·hit_outs + 0.5·one_percenters - 1.5·clangers - 1.0·free_kicks_against

Design intent. Goals weigh 4× disposals so MIS does not collapse into "whoever touched it most"; negative weights rank a high-turnover game below an efficient one; rebound_50s, one-percenters and hit_outs let defenders and ruckmen compete. **Validated against real Brownlow votes (2003–2024): the MIS leader polls votes in ~61% of matches versus a ~7% random baseline — a ~9× lift — beating both AFL Fantasy and raw disposals.** The three top-player targets select near-disjoint populations (midfielders → disposals, key forwards → goals, MIS admits defenders), so **build a separate model per target.**

4 · Feature families — 109 numeric features (full detail: feature_dictionary_v2.0.0.csv)

Every team feature is stored three times: home_*, away_*, diff_* (home – away). The difference form is usually most informative — in a two-sided contest the gap matters more than either level.

Family	Features	Computation window	Leakage control
Strength	elo_pre, elo_diff, elo_exp_home_win	all history, 25%/season regression	rating recorded before update
Form	win rate, avg score for/against, avg margin, margin sd	last 3 / 5 / 10 matches	.shift(1) before .rolling()
Momentum	win_streak (signed)	current run	computed on shifted series
Season	ssn_points, ssn_percentage, ssn_win_rate, ladder_pos	season to date	.shift(1) before .cumsum()
Head-to-head	win rate & avg margin vs this opponent	last 10 meetings + all-time	.shift(1) before rolling
Context	days_rest, is_interstate, is_travelling, venue_win_rate_prior	varies / static	prior matches only
Player form	pform_{disp,goals,fp,mis}_avg_{3,5}, pform_career_disp_avg	last 3 / 5 games	.shift(1) before .rolling()

Excluded deliberately: crowd (a genuine figure here, but known only after the match), final and quarter scores, and every player statistic from the match itself. **Weather is unavailable** — no source file carries temperature, rainfall, wind or roof state; rest days, interstate travel and venue familiarity stand in for the physical-context effects it would proxy. **Positions are unavailable** — role is derived from career statistical profile and labelled a proxy throughout (64% fall into a General/Utility catch-all).

5 · Raw-data defects found and repaired.

#	Defect	Severity	Repair
1	Team names in 40 (match) / 114 (seasonal) variants — tabs, padding, lowercase, "W. Bulldogs"	CRITICAL	Case-insensitive canonical map. H/A pairing 50.5% → 100%
2	Venue names in 64 variants	high	Canonical map → 37 grounds + state
3	margin disagrees with score difference	high	Recomputed from scores; never trusted
4	Exact duplicate rows (10 + 10 + 5) and duplicate player id	medium	Dropped / kept-first
5	889 negative counting statistics; weight = 0; negative games played	high	Set to NaN — physically impossible
6	score column 100% null	low	Column dropped
7	Blanks in counting stats mean ZERO, not missing (min observed = 1.0 for every stat)	CRITICAL	Era-aware zero fill; corrects ~3× bias (goals 1.6 → 0.53/game)
8	No match_id or team_id in any file	high	Both derived; 0.000% unlinked
9	Round 0 ("Opening Round", 2024+)	low	Leakage test keyed on each season's minimum round

6 · Train / hold-out split and realistic ceiling

Split	Seasons	Matches	Note
TRAIN	1997–2021	4,811	Window tested empirically; 1983–96 seasons add little
VAL	2022–2023	423	Hyper-parameter selection only

TEST (hold-out)	2024	216	Most recent COMPLETE season; 2025 truncated
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Why not a random split. (i) It lets round 22 of 2019 sit in train while round 3 of 2019 sits in test — predicting an early-season match having already seen how the season played out. (ii) Team strength is strongly autocorrelated, so matches a fortnight apart are near-duplicate observations. (iii) The competition is non-stationary (12→18 teams, drifting scoring rates and home advantage), so a random split never asks the model to extrapolate forward into an unseen season. **Measured, not asserted: a single un-shifted rolling mean inflated hold-out accuracy from 0.620 → 0.954; a random split inflated it to 0.676.**

Realistic ceiling. AFL outcomes are irreducibly noisy: 47% of the hold-out season was decided by four goals or fewer, and a single injury, late umpiring decision or wet afternoon can flip a result no pre-match information could anticipate. The practical ceiling is about 70–75% — bookmakers, pricing with far richer information and financially punished for error, land near 72–74%. A realistic Day-2 target is 68–72%, against an always-tip-home baseline of 0.569. **If a model reports 85%, 95% or 99% accuracy, that is a bug, not a breakthrough** — near-certain evidence of leakage: a rolling feature missing .shift(1), a shuffled split, a post-match statistic left in, or the target surviving as an input. Diagnostic: the model scores 44.7% on matches decided by ≤ 6 points (a coin flip) and 75.4% on 40+ point margins — the information simply is not there for close games.

7 · Day-2 contract

1. Load data/features/match_features_v2.0.0.parquet and verify content_md5 against manifest_v2.0.0.json. Do not rebuild features ad hoc.
2. Call time_split() from src/afl_features.py. Never train_test_split(shuffle=True).
3. Targets are home_win (primary) and margin (secondary). Never feed home_score, away_score, margin, result or total_score in as inputs.
4. Player targets: is_top_mis (primary), is_top_disposal_getter, is_top_goal_kicker — a separate model each.
5. **If hold-out accuracy exceeds ~0.78, stop and audit for leakage before reporting anything.**

Baselines to beat on the held-out 2024 season

Model	Accuracy	AUC	Brier
Always tip the home team	0.569	—	0.245
Higher-ranked team on the ladder	0.593	—	—
Elo probability alone	0.657	0.716	0.210
Logistic regression, full clean feature set	0.620	0.692	0.215